

David Khachatryan - Week 5 Report

Summary

For Milestone 1, as the zoomed in picture shows, the filtered pitch (orange line) is able to keep up with the desired pitch (blue line), well before the saturation limit (purple) is reached. When adding I gains, it was relatively straightforward to get the error to a minimal number. It might still need to be adjusted further.

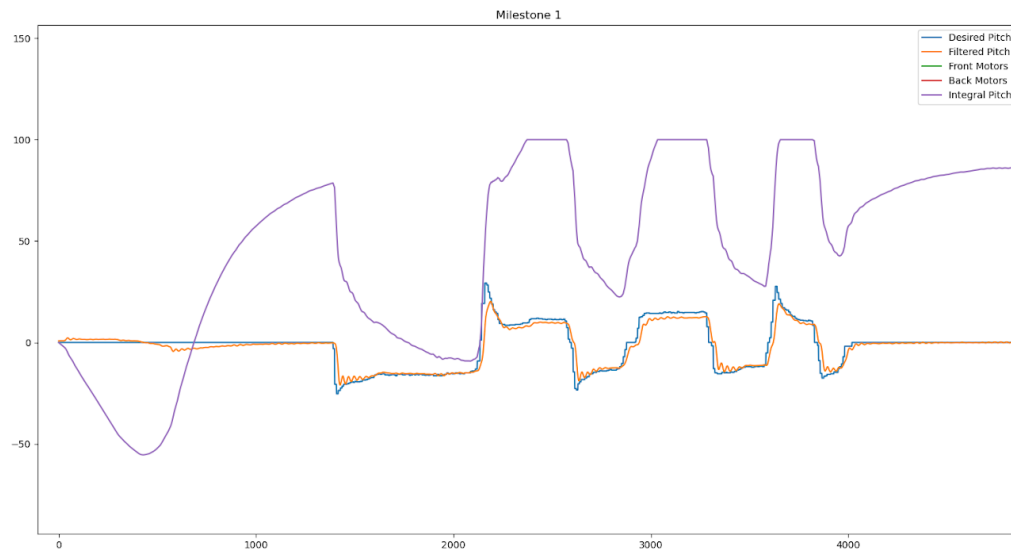


Figure 1: Milestone 1 Zoomed in



Figure 2: Milestone 1

For milestone 2, desired roll was calculated using the same amplitude. Implementing the same PID controller achieved the control graph below where error is minimal. Roll term was calculated to be:

```
float roll_term = ROLL_P_GAIN * r_error
                - ROLL_D_GAIN * imu_data[4]
                + integral_roll;
```

The pitch term for the right side negative motors were negative and for the left side motors were positive.

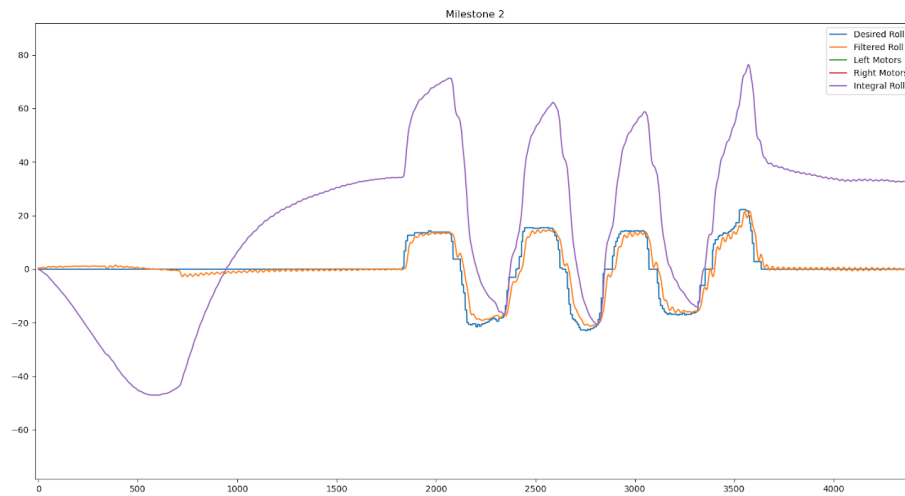


Figure 3: Milestone 2 Zoomed in



Figure 4: Milestone 2

What went well

Adding the pause motor command went well. I know other groups had the issue of a motor beeping when paused, but this did not happen to me. I was given a hint that instead of setting the values to 0 they should be set to 1. Not sure why this fixed it, but this was not something I spent a lot of time on, which was great.

What did not go well, why?

In the code, I originally wrote the pitch and roll formula as one equation. This created a lot of issues and made it impossible to debug milestone 2, when roll was the only thing to be used for motor command calculations.

In order to fix the problem, I rewrote the motor command as $\text{thrust} \pm \text{pitch_term} \pm \text{roll_term}$ and calculated each term respectively. This made it easier for me to figure out the signs and turn the right motors in the aftermath of joystick inputs.

What will you change for next class

For yaw control, it would be easier to separate the yaw term when counting motor commands. Another thing that needs to be done is further tuning. The robot was oscillating during roll control, which was further confirmed by my graph. For the purpose of real-time flight, further tuning would be necessary for the next class.

Contributions

This week Burke and I worked separately. I finished milestones 1 and 2 during class and milestone 3 shortly after the class. So, numerically I we each did 100% of the work separately